

Topics in Applied Graph Theory

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based on course by Priyavrat Deshpande

Evaluation: Assignments + Projects

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1. Introduction

Definition: Graph

A graph $G = (V, E)$ is specified by two finite sets V and E where V is the set of vertices and E is the set of edges (a subset of 2^V).

Definition: Directed Graph

A directed graph $G = (V, E)$ is specified by two finite sets V and E where V is the set of vertices and E is the set of edges (a subset of $V \times V$ barring (x, x) or loops).

Side Note

The graph is simple and undirected (notice 2^V subsets) while a directed graph is, well, directed (notice the cartesian product).

Definition: Multigraph

A multigraph G has E be a multiset.

Definition: Weighted

We say a graph is (edge) weighted, if we have a function $w : E \rightarrow \mathbb{R}$

For notational convenience, we take $V = [n]$.

Definition: Adjacency Matrix

Adjacency matrix of graph G , called A_G is a $n \times n$ matrix where $a_{i,j} = 1 \iff (i, j) \in E$ and 0 otherwise.

In weighted graph, $a_{i,j} = w((i, j))$

Definition: Degree

The degree of vertex i is number of edges containing said vertex, that is $d_G(i) = \deg(i) = \sum_{j=1}^n a_{i,j}$.

Definition: Degree Matrix

The degree matrix of D is a diagonal matrix $D(i, i) = d(i)$.

Definition: Laplacian

The Laplacian of graph G is $\mathcal{L} = D - A$

Definition: Graph Energy

For $x \in \mathbb{R}^n$, $x^T \mathcal{L} x$ is called graph energy.

Example

Given a 2 path, $A = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$, $D = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \Rightarrow \mathcal{L} = \begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix} \Rightarrow x^T \mathcal{L} x = (x_1 - x_2)^2 \geq 0$.

Similarly, for a 3 path, $x^T \mathcal{L} x = (x_1 - x_2)^2 + (x_2 - x_3)^2$

Theorem 1.1. For a weighted graph,

$$x^T \mathcal{L} x = \sum_{(i,j) \in E} w_{i,j} (x_i - x_j)^2$$

Proof.

$$\begin{aligned}
x^T \mathcal{L}x &= x^T (D - W)x \\
&= x^T Dx - x^T Wx \\
&= \sum d_i x_i^2 - \left(\sum w_{i,j} x_i x_j \right) \\
&= \sum d_i x_i^2 - 2 \sum_{(i,j) \in E} w_{i,j} x_i x_j \\
&= \sum_{(i,j) \in E} (w_{i,j} x_i^2 - 2w_{i,j} x_i x_j + w_{i,j} x_j^2) \\
&= \sum_{(i,j) \in E} w_{i,j} (x_i - x_j)^2
\end{aligned}$$

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Corollary 1.2. $\mathcal{L}$ is Symmetric and Positive Semi-Definite.

This implies that $\mathcal{L}$ has non-negative real Eigenvalues.

Definition: Spectrum of Laplacian

We define $\sigma(\mathcal{L}) := \{0 \leq \lambda_1 \leq \lambda_2 \leq \dots \leq \lambda_n\}$ as the spectrum of $\mathcal{L}$

Let $\mathbf{1} = (1, 1, \dots, 1)^T$. Notice,

$$\begin{aligned}
\mathcal{L}\mathbf{1} &= (D - W) \begin{pmatrix} 1 \\ \vdots \\ 1 \end{pmatrix} \\
&= D \begin{pmatrix} 1 \\ \vdots \\ 1 \end{pmatrix} - W \begin{pmatrix} 1 \\ \vdots \\ 1 \end{pmatrix} \\
&= \begin{pmatrix} d_1 \\ \dots \\ d_n \end{pmatrix} - \begin{pmatrix} \sum w_{1,j} \\ \dots \\ \sum w_{n,j} \end{pmatrix} \\
&= 0
\end{aligned}$$

which implies $\lambda_1(\mathcal{L}) = 0$ with eigenvector $\mathbf{1}$.

Theorem 1.3. $\lambda_2 > 0 \iff G$ is connected

Proof. ($\implies$) Assume $G = G_1 \sqcup G_2$ which implies $\mathcal{L}_G = \begin{pmatrix} \mathcal{L}_{G_1} & 0 \\ 0 & \mathcal{L}_{G_2} \end{pmatrix}$.

This implies we have atleast 2 independent vectors: $\begin{pmatrix} \mathbf{1}_{G_1} \\ 0 \end{pmatrix}$ and $\begin{pmatrix} 0 \\ \mathbf{1}_{G_2} \end{pmatrix}$ which have eigenvalue 0 which makes $\lambda_2 = 0$.

($\impliedby$) Suppose G is connected. Let $y \in \ker(\mathcal{L}_G)$.

$$\begin{aligned}\Rightarrow y^T \mathcal{L}y &= \sum_{(i,j) \in E} (x_i - x_j)^2 \\ &= 0\end{aligned}$$

This implies that $x_i = x_j \iff (i, j) \in E$. Thus, if vertex i and j are connected, then by a series of equalities, we'll end up with $y = \mathbb{1}$.

Thus, $\lambda_2 > 0$. ■